

Dominic DeCarlo

412-773-9883 | dominic2decarlo@gmail.com | [linkedin.com/in/domdecarlo](https://www.linkedin.com/in/domdecarlo) | github.com/dominicdecarlo | dominicdecarlo.vercel.app

EDUCATION

University of Pittsburgh

Pittsburgh, PA

Bachelor of Science in Computer Science and Data Science

August 2023 – April 2027

Relevant Coursework: Data Structures & Algorithms, Intro to AI, Machine Learning, Software Engineering, Data Science, Systems Software, Discrete Math, Calculus, Linear Algebra, Formal Methods, Stats, Probability

PROJECTS

Ground-Up Large Language Model (LLM) | *Python, PyTorch*

April 2026

- Architected a custom transformer-based neural network from the ground up, executing distributed pre-training pipelines across high-performance A100/H100 GPU clusters to optimize the model architecture
- Fine-tuned conversational behavior using a curated proprietary dataset and engineered a stable deployment pipeline to serve the optimized model weights via a live, web-hosted interface.

Fine-Tuned Local AI Coding & Voice Assistant | *PyTorch, Unsloth, QLoRA*

August 2026

- Fine-tuned a 9B parameter LLM via QLoRA to optimize code generation and reasoning, surpassing the larger Qwen 3.5 27B model on 7 standard coding benchmarks.
- Deployed a fast, local voice agent using STT & TTS pipelines with an LM Studio backend for live hands-free use.

AI Soccer Analysis System | *Python, OpenCV, YOLO*

July 2024

- Developed an AI/ML system using Python and computer vision to analyze football/soccer matches.
- Implemented real-time object detection to analyze player movement, important game events, etc.
- Provided valuable insights to improve match evaluation and strategy

College Real Estate Predictor | *Python, Numpy, Sci-Kit Learn, Pandas*

October 2024

- Created a machine learning model to provide more accurate current and future housing value and rent predictions than Zillow estimates, tailored for landlords and investors.
- Merged and gathered student and property data using Pandas for analysis and predictions

Real-Time Prediction Market Arbitrage SaaS | *Python, FastAPI, Google Stitch*

February 2026

- Engineered a Python scanner to instantly detect profitable arbitrage bets across Kalshi and Polymarket
- Built an interactive dashboard to sort live market spreads by profit margin with direct execution links to exchanges.

EXPERIENCE

AI & Automation Intern

May 2026 – December 2026

Thermo Fisher Scientific

Pittsburgh, PA

- Engineered an ensemble ML model predicting customer experience scores and root-cause risk factors for IT incident tickets, saving managers 98% of manual triage time and projecting ~\$110k in operational savings.
- Assessed frontier model vulnerabilities and developed AI security frameworks to enable the safe deployment of high-performing LLMs.

IT Intern, Data Insights and Reporting Team

May 2025 – August 2025

National Philanthropic Trust

Philadelphia, PA

- Built a Python-based AI/ML model to forecast contribution trends, helping the team proactively strategize for the high-volume giving season.
- Performed data analysis on complex datasets to develop business-focused data visualizations (Sankey charts, Power BI dashboards), improving reporting efficiency and providing actionable insights into donor behavior.

CAMPUS INVOLVEMENT

Pitt Computer Science Club

August 2024 – Present

- Contributed to collaborative projects and technical discussions, boosting problem-solving and technical knowledge.

Panther Equity

January 2025 – Present

- Gained hands-on finance experience through market trend analysis and equity research for simulated portfolios.

TECHNICAL SKILLS

Languages: Java, Python, R, SQL, C, C++, Go, HTML/CSS, JavaScript, TypeScript, Bash, Assembly

Frameworks/Libraries: PyTorch, TensorFlow, Scikit-Learn, NumPy, React, Svelte, Node.js, OpenCV

Developer Tools: GitHub, Git, Jira, n8n, Power BI, Markdown, Visual Studio Code, Stitch, Excel